

Course: B. Tech.

Branch: Civil Engineering

Semester: VI

Subject Code & Name: BTCVC603\_Y22 Transportation Engineering

Max Marks: 60

Date: 19/6/2024

Duration: 3 Hr.

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                                                                                                                                                                                                        | (Level/CO) | Marks     |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                 |            | <b>12</b> |
| A)          | Write an explanatory note on Highway Planning Survey's                                                                                                                                                                                                                                                 | CO1        | 6         |
| B)          | State and explain different types of road pattern with diagram.                                                                                                                                                                                                                                        | CO1        | 6         |
| C)          | State and explain classification of road based on materials with diagram.                                                                                                                                                                                                                              | CO1        | 6         |
| <b>Q.2</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                 |            | <b>12</b> |
| A)          | Define: a) Right of way b) Road Margin c) Carriage way<br>d) Shoulder e) Kerb f) Camber                                                                                                                                                                                                                | CO 2       | 6         |
| B)          | The speed of overtaking and overtaken vehicles are 70 & 40 kmph respectively on two way traffic road. If the acceleration of overtaking vehicle is $0.99 \text{ m/s}^2$ .<br>(i) Calculate safe overtaking sight distance.<br>(ii) Mention the minimum length overtaking zone (Assume $t = 2$ Seconds) | CO 2       | 6         |
| C)          | Explain overtaking sight distance in detail with diagram.                                                                                                                                                                                                                                              | CO 2       | 6         |
| <b>Q. 3</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                 |            | <b>12</b> |
| A)          | State different types of tests carried in the laboratory on the sample of aggregate to decide the suitability for its use in road construction and explain any one of them in detail.                                                                                                                  | CO 3       | 6         |
| B)          | Define : a) Bitumen b) Asphalt c) Tar d) Emulsion e) Cutback f) Road Oil                                                                                                                                                                                                                               | CO 3       | 6         |
| C)          | In the Marshall method of mix design the coarse aggregate, fine aggregate, filler and bitumen having respective specific gravities of 2.62, 2.72, 2.70 & 1.02 are mixed in the ratio 55, 34.6, 4.8 and 5.6 percent respectively. What will be the theoretical specific gravity of the mix?             | CO 3       | 6         |

**Q.4 Solve Any Two of the following.**

- |    |                                                                           |      |   |
|----|---------------------------------------------------------------------------|------|---|
| A) | Write a note on Speed Studies.                                            | CO 4 | 6 |
| B) | State and explain the methods for collecting origin and destination data. | CO 4 | 6 |
| C) | What is Traffic Engineering? Explain traffic characteristics.             | CO 4 | 6 |

**Q. 5 Solve Any Two of the following. 12**

- |    |                                                                 |      |   |
|----|-----------------------------------------------------------------|------|---|
| A) | Explain different components of pavement with its function?     | CO 5 | 6 |
| B) | Explain the advantages and limitations of CBR method of design. | CO 5 | 6 |
| C) | Explain types of joints in Cement concrete road.                | CO 5 | 6 |

**\*\*\* End \*\*\***